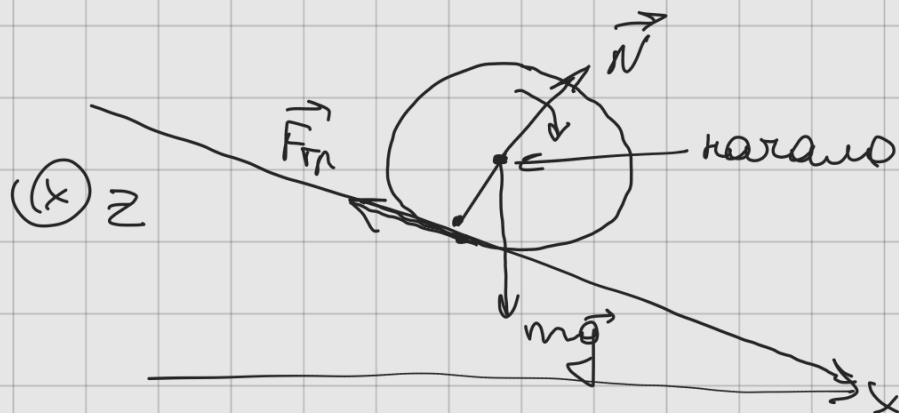


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Решение:



$$L_z = L_{y, M, z} + L_{OTH. y, M, z} = 0 + I\omega = I\omega, \quad I = \frac{mR^2}{2}$$

$$M_z = R \cdot F_{тр}$$

$$\bullet \quad \frac{dL_z}{dt} = M_z \Rightarrow R \cdot F_{тр} = I \frac{d\omega}{dt}$$

$$\bullet \quad \text{Условие качения: } v = \omega R \Rightarrow a = \frac{d\omega}{dt} R$$

$$R \cdot F_{тр} = I \cdot \frac{a}{R}$$

$$\bullet \quad \text{II закон н. н. х: } mg \sin \alpha - \mu mg \cos \alpha = ma$$

$$mg \sin \alpha - F_{тр} = ma$$

$$a = g \sin \alpha - \frac{F_{тр}}{m}$$

$$R^2 \cdot F_{тр} = I \cdot (g \sin \alpha - \frac{F_{тр}}{m})$$

$$(R^2 + \frac{I}{m}) F_r = I g \sin \alpha$$

$$F_r = \frac{I g \sin \alpha}{R^2 + \frac{I}{m}} = \frac{\frac{m R^2}{2} g \sin \alpha}{R^2 + \frac{R^2}{2}} = \frac{\frac{m}{2} g \sin \alpha}{\frac{3}{2}} = \frac{m g \sin \alpha}{3}$$

$$m g R \sin \alpha = I \frac{a}{R} + m a R$$

$$m g \sin \alpha - F_r = m a$$

$$m g R \sin \alpha = \frac{I}{R} \cdot \left(\frac{m g \sin \alpha - F_r}{m} \right) + m R \left(g \sin \alpha - \frac{F_r}{m} \right)$$

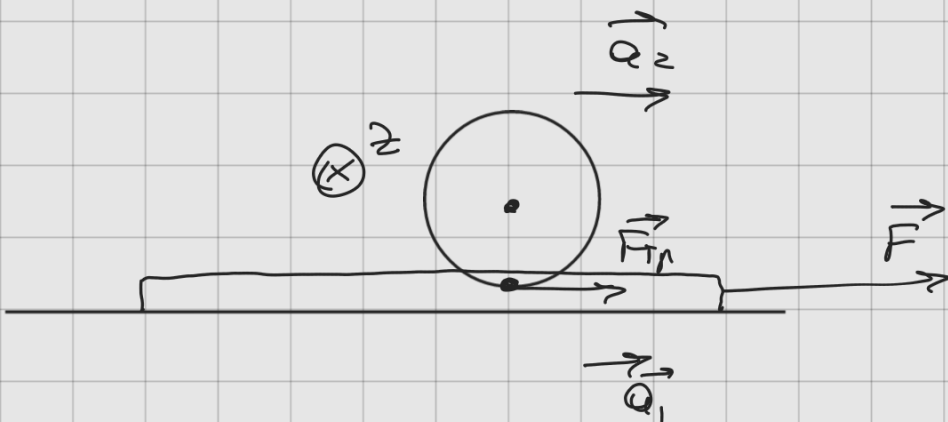
$$\cancel{m g R \sin \alpha} = \cancel{\frac{m R}{2}} \left(g \sin \alpha - \frac{F_r}{m} \right) + \cancel{m R} \left(g \sin \alpha - \frac{F_r}{m} \right)$$

$$\cancel{g \sin \alpha} = \frac{g \sin \alpha}{2} - \frac{F_r}{2m} + \cancel{g \sin \alpha} - \frac{F_r}{m}$$

$$+ \frac{\cancel{g \sin \alpha}}{2} = + \frac{3}{2} \frac{F_r}{m} \rightarrow F_r = \frac{m g \sin \alpha}{3}$$

N51

Peuerme:



$$1) F = m_1 a_1 + m_2 a_2$$

$$2) F_{TP} = m_2 a_2 \rightarrow a_2 = \frac{F_{TP}}{m_2}$$

$$3) M_z = \frac{dL_z}{dt}, L_z = \underbrace{L_{y.M}}_0 + L_{\text{rot. y.M}}$$

$$L_{\text{rot. y.M}} = I \omega \Rightarrow \frac{dL_z}{dt} = I \frac{d\omega}{dt}$$

$$a_2 = a_1 - a_0$$

$$F_{TP} \cancel{R} = I \frac{a_0}{R} \Rightarrow \frac{2m_1 \cancel{R}^2 \cdot a_0}{\cancel{R}} \quad Q_0 = \frac{5F_{TP}}{2m_2}$$

$$a_1 = \frac{7 F_{TP}}{2 m_2}$$

$$F = \frac{7 F_{TP} m_1}{2 m_2} + m_2 \cdot \frac{F_{TP}}{m_2} = \frac{7 F_{TP} m_1 + 2 m_2 F_{TP}}{2 m_2}$$

$$F - F_{TP} = m_1 a_1$$

$$F = F_{TP} + \frac{7 m_1 F_{TP}}{2 m_2} \Rightarrow F_{TP} = \frac{F}{1 + \frac{7 m_1}{2 m_2}} = \frac{2 F m_2}{2 m_2 + 7 m_1}$$

$$F = F_{TP} + m_1 \cdot \frac{7 F_{TP}}{2 m_2}$$

$$F = m_1 \cdot \frac{7 F_{TP}}{2 m_2} + \frac{m_2 \cdot F_{TP}}{m_2}$$

$$F - F_{TP} = m_1 a_1 \Rightarrow F_{TP} = F - m_1 a_1$$

$$F - F_{TP} = m_1 \cdot \frac{7 F_{TP}}{2 m_2}$$

$$F_{TP} = \frac{F}{1 + \frac{7 m_1}{2 m_2}}$$

$$F = m_1 a_1 + m_2 a_2 \quad a_2 = a_1 - a_0$$

$$F - F_{TP} = m_1 a_1$$

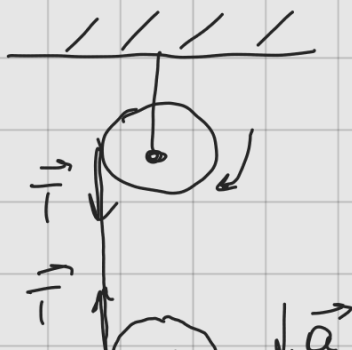
$$F_{TP} = m_2 a_2 \rightarrow a_2 = \frac{F_{TP}}{m_2}$$

$$F_T \cdot R = I \cdot \frac{a_0}{R} \rightarrow a_0 = \frac{F_T R^2}{I} = \frac{F_T \cdot R^2}{\frac{2}{5} m_2 R^2} = \frac{5 F_T}{2 m_2}$$

$$a_1 = \frac{7 F}{2 m_2 \cdot \left(1 + \frac{7 m_1}{2 m_2}\right)} = \frac{7 F}{2 m_2 + 7 m_1}$$

$$a_2 = \frac{F_{TP}}{2 m_2} = \frac{F}{m_2 \left(1 + \frac{7 m_1}{2 m_2}\right)} = \frac{F}{m_2 + \frac{7 m_1}{2}}$$

N52



$$T - mg = -ma$$

$$T \cdot R = \frac{m R^2}{2} \cdot \frac{a_1}{R}$$

$$T = \frac{m a_1}{2}$$



$$1 - R = \frac{mR}{2} \cdot \frac{a_2}{R}$$

$$T = \frac{ma_1}{2} \Rightarrow a_1 = \frac{2T}{m}$$

$$a_2 = \frac{2T}{m}$$

$$a = a_1 + a_2 = \frac{4T}{m} \Rightarrow T - mg = -4T$$

$$5T = mg$$

$$T = \frac{mg}{5}$$

$$a = \frac{4}{5}g$$